
MySQL – JOINS (Full Notes Book Format)

Step 1: Create Sample Tables

```
(104, 'C++');
```

Data in `students` Table:

<code>student_id</code>	<code>name</code>	<code>course_id</code>
1	Ankit	101
2	Riya	102
3	Aman	101
4	Sneha	103
5	Rohit	NULL

Data in `courses` Table:

<code>course_id</code>	<code>course_name</code>
101	Python
102	Java
103	SQL
104	C++

----->

1. INNER JOIN

```
SELECT students.name, courses.course_name
FROM students
INNER JOIN courses ON students.course_id = courses.course_id;
```

☞ Sirf wo rows milengi jinke dono tables me matching `course_id` ho.

☞ Rohit (NULL) aur `course_id` 104 (unused) exclude ho jaayenge.

Output:

<code>name</code>	<code>course_name</code>
Ankit	Python

name course_name

Riya Java

Aman Python

Sneha SQL

----->

✓ 2. LEFT JOIN

```
SELECT students.name, courses.course_name
FROM students
LEFT JOIN courses ON students.course_id = courses.course_id;
```

☞ Sabhi students milenge, chahe unka course match ho ya na ho.

☞ Rohit bhi dikhai dega, jiska course_id NULL hai.

📤 Output:

name course_name

Ankit Python

Riya Java

Aman Python

Sneha SQL

Rohit NULL

----->

✓ 3. RIGHT JOIN

```
SELECT students.name, courses.course_name
FROM students
RIGHT JOIN courses ON students.course_id = courses.course_id;
```

☞ Sabhi courses dikhte hain, chahe student assigned ho ya nahi.

☞ C++ bhi dikhai dega, jisme koi student nahi hai.

📤 Output:

name course_name

Ankit Python

Riya Java

Aman Python

name	course_name
Sneha	SQL
NULL	C++

----->

✔ 4. FULL OUTER JOIN (MySQL me directly nahi hota — workaround se hota hai)

```
SELECT s.name, c.course_name
FROM students s
LEFT JOIN courses c ON s.course_id = c.course_id
UNION
SELECT s.name, c.course_name
FROM students s
RIGHT JOIN courses c ON s.course_id = c.course_id;
```

☞ Dono tables ke sabhi rows milte hain, chahe match ho ya na ho.

☞ Rohit aur C++ dono included honge.

📤 Output:

name	course_name
Ankit	Python
Riya	Java
Aman	Python
Sneha	SQL
Rohit	NULL
NULL	C++

----->

✔ 5. SELF JOIN (Table joins with itself)

```
SELECT A.name AS Student1, B.name AS Student2
FROM students A, students B
WHERE A.course_id = B.course_id AND A.student_id <> B.student_id;
```

☞ Ek hi table ke students compare ho rahe hain jo same course me hain.

☞ Same course me multiple students ho to unko pair bana ke dikhata hai.

📤 Output:

Student1 Student2

Ankit Aman

Aman Ankit

----->

✓ Summary Table – Types of JOINS

JOIN Type	Description
INNER JOIN	Only matching rows from both tables
LEFT JOIN	All from left + matching from right
RIGHT JOIN	All from right + matching from left
FULL OUTER JOIN	All rows from both (via UNION in MySQL)
SELF JOIN	Table joins with itself
